

Computing CMB temperature fluctuations for spherical spaces

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Manifolds & Homotopy Groups

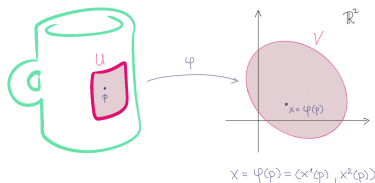


Figure: The prototypical example of a manifold a mug [13].

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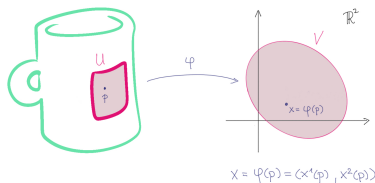


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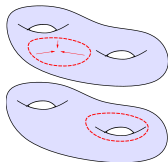


Figure: Two double tori with (non)-contractible paths [7].

Quotients of the 3-sphere

- $SO(4)$ is isomorphic to the isometry group of $\mathbb{S}^3$.
- Subgroups $\Gamma \leq SO(4)$ define **equivalence classes** of orbits on $\mathbb{S}^3$ by the standard action of $SO(4)$ on $\mathbb{R}^4$, i.e. $x \sim y$ iff $x = My$ for some $M \in \Gamma$.

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- The obtained space is (*sometimes*) a manifold. In particular, it is well defined and spherical for finite Γ .
- This can be easily generalized to the n -sphere.

Lens Spaces — the Explicit Construction

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Definition (Lens space)

Given coprime $p, z \in \mathbb{Z}$, we define

$$L(p, q) := \mathbb{S}^3 / \langle M \rangle,$$

where $\langle M \rangle$ is the group generated by

$$M := \begin{bmatrix} \cos(2\pi/p) & \sin(-2\pi/p) & 0 & 0 \\ \sin(2\pi/p) & \cos(2\pi/p) & 0 & 0 \\ 0 & 0 & \cos(2\pi q/p) & \sin(-2\pi q/p) \\ 0 & 0 & \sin(2\pi q/p) & \cos(2\pi q/p) \end{bmatrix}.$$

The Laplace-Beltrami operator

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a generalization of the Laplace operator to more general spaces

$$\Delta f := \operatorname{div} \nabla f.$$

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- 2 The spectrum determines many things about the space (like its volume).
- 3 When two distinct (up to isomorphism) manifolds share a spectrum, they form an *isospectral pair*.

The silver lining

- Isospectral pairs are intriguing in their own right, but they cannot occur in dimension 3.
- We can thus attempt to infer the shape of our universe based on its spectrum.

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- The set of all possible wave numbers [for which a nonzero solution exists] depends on Γ - usually a proper subset of $\mathbb{N}$.
- Solutions to the Helmholtz equation are calculated for each Γ and CMB anisotropy $\frac{\delta T}{T}$ computed as a sum of spherical harmonics.

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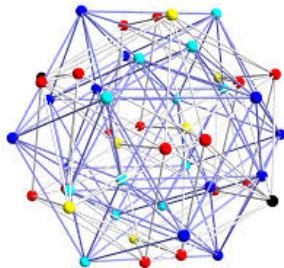


Figure: Graphical representation of the binary tetrahedral group [5]

Inhomogeneous spherical space

- Manifolds of the form $\mathbb{S}^3/\Gamma$ with Γ a group acting on $\mathbb{S}^3$.
- **Multi-connected** space: it has non-contractable loops.
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 - 1 homogeneous: $N3$ and $L(8, 1)$.
 - 2 inhomogeneous: $N2 \equiv L(8, 3)$.
- Results: inhomogeneous spaces have **more variety** in the CMB anisotropies than homogeneous spaces, because the strength of anisotropy suppression is **observer dependent**.

Statistical Isotropy and Hypothesis

Theoretical Expectation:

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Key Question: - Does the observed CMB data deviate from statistical isotropy? - If $\langle a_{\ell m} \rangle \neq 0$, this suggests a preferred cosmic direction.

Methodology: Sky Masking and the Test Statistic

Challenges in Real Observations:

- We cannot observe the full CMB sky due to foreground contamination.
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Monte Carlo Simulations:

- We generate thousands of **simulated CMB skies** assuming isotropy.
- Each sky has different random $a_{\ell m}$, drawn from:

$$a_{\ell m} \sim \mathcal{N}(0, C_\ell)$$

- The observed WMAP data is compared against these simulations.

Results and Interpretation

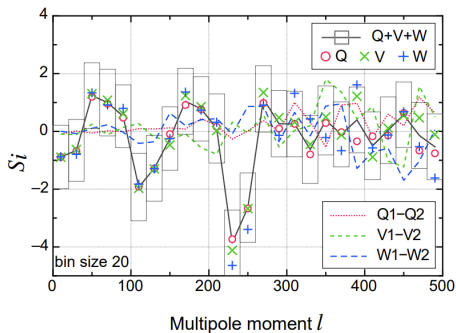
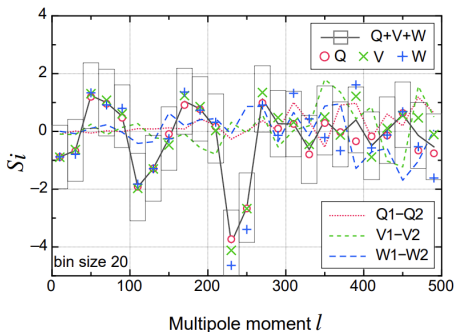


Figure: The decorrelated band mean test statistic values over multipole ranges [9].

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Key Findings:

- A statistically significant deviation was found in $221 \leq \ell \leq 240$.
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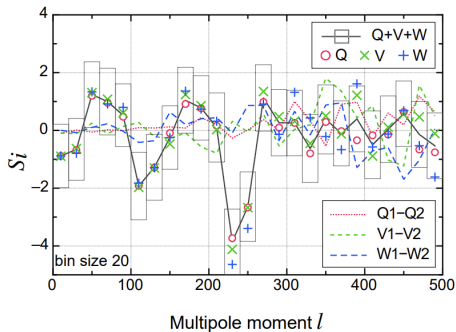


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Possible Explanations:

- A real cosmological signal? → A finite universe or new physics.
- A systematic effect? → Foreground contamination or instrumental noise.

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- ② There are two homogeneous spherical manifolds obtained as $\mathbb{S}^3/\Gamma$ which produce CMB similar to observations.
- ③ Inhomogeneous spherical spaces exhibit varied behavior of CMB anisotropies.
- ④ Statistical test results suggest possibilities of **finite multi-connected** topology.

References

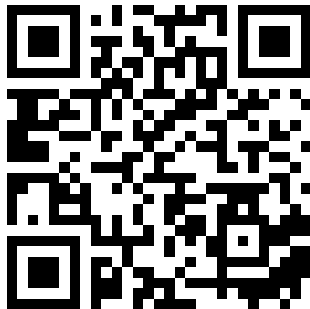
- 1 R. Aurich, P. Kramer, and S. Lustig. *Cosmic microwave background radiation in an inhomogeneous spherical space*. Physica Scripta, 2011.
- 2 R. Aurich, P. Kramer, and S. Lustig. *A survey of lens spaces and large-scale CMB anisotropy*. Monthly Notices of the Royal Astronomical Society, 2012.
- 3 R. Aurich, S. Lustig, and F. Steiner. *CMB anisotropy of spherical spaces*. Classical and Quantum Gravity, 2005.
- 4 P. Bielewicz, K. M. Górski, and A. J. Banday. *Low-order multipole maps of cosmic microwave background anisotropy derived from WMAP*. Oxford University Press, 2004.
- 5 G. Egan. *Symmetries and the 24-cell*. 2021. Accessed at <https://www.gregegan.net/SCIENCE/24-cell/24-cell.html>.

References 2 — Electric Boogaloo

- 6 N. Jarosik et. al. *Seven-year Wilkinson Microwave Anisotropy Probe (WMAP) Observations: Sky Maps, Systematic Errors, and Basic Results*. The American Astronomical Society, 2011.
- 7 M. Herlihy and N. Shavit. *The Topological Structure of Asynchronous Computability*. Journal of the ACM, 1996.
- 8 A. Ikeda. *On the spectrum of a Riemannian manifold of positive constant curvature*. Osaka Journal of Mathematics, 1980.
- 9 D. Kashino, K. Ichiki, and T. Takeuchi. *Test for anisotropy in the mean of the CMB temperature fluctuation in spherical harmonic space*. Physics Review, 2012.
- 10 P. Labrana. *Emergent Universe Scenario and the Low CMB Multipoles*. Physical Review, 2013.

References 3 — the References Strike Again

- 11 E. A. Lauret and B. Linowitz. *The spectral geometry of hyperbolic and spherical manifolds: analogies and open problems*. New York Journal of Mathematics, 2025.
- 12 R. Lehoucq, J. Weeks, J. P. Uzan, E. Gausmann, and J.P. Luminet. *Eigenmodes of three-dimensional spherical spaces and their application to cosmology*. Classical and Quantum Gravity, 2002.
- 13 M. Serri. *Analysis on Manifolds*. AMS Open Math Notes, 2025.
- 14 B. Tai-Dana, T. Bryson, and J. Terilla. *Topology, a categorical approach*. The MIT Press, 2020.



💫 Thank You! 💫